Handout for Week 8—Frege in Context

Course materials available at: https://spaceofreasons.netlify.app/courses/frege2025/

Plan: Using logical tools to elaborate *old* concepts-and-claims into *new* ones—
in different senses of 'new':

1. Boolean combinations of microfeatures. Iterate and build to the sky.
2. First bridge: Monadic—>polyadic simple predicates.
3. Second bridge: Functional analysis: simple predicates —> complex predicates.

Quantifiers taking complex predicates to sentences.

4. Cycle of composition and decomposition. Iterate and build to the sky.
5. Third bridge: single premise to multipremise implications.

Alternating quantifiers. Context of the point in Kant and Euclid.

6. Comparing Euclidean roles with Fregean senses as constellations of modes of presentations (functions-applied-to-arguments) that determine patterns of inference.
7. Fourth bridge: Abstraction as introducing new sortal concepts and referring terms.

Deduction-Detachment (DD): $\Gamma, A \mid\sim B, \Delta$

========= Bidirectional Meta-Inference Line

$\Gamma \mid\sim A \rightarrow B, \Delta$.

Incoherence-Incompatibility (II): $\Gamma, A \mid\sim \Delta$

========= Bidirectional Meta-Inference Line

$\Gamma \mid\sim \neg A, \Delta$.

Concepts-and-claims formed by complex predicates can be thought of as formed by a four-stage process.

• First, put together simple predicates and singular terms, to form a set of sentences, say {Rab,Sbc,Tacd}.

• Then apply sentential **compounding** operators to form more complex sentences, say {Rab→Sbc, Sbc&Tacd}.

• Then **analyze** by discerning functions by substituting variables for some of the singular terms (individual constants), to form complex predicates, say {Rax→Sxy, Sxy&Tayz}.

• Finally, apply quantifiers to bind some of these variables, to form new sentences and complex predicates, for instance the one-place predicates (in y and z) {∃x[Rax→Sxy], ∀x∃y[Sxy&Tayz]}.

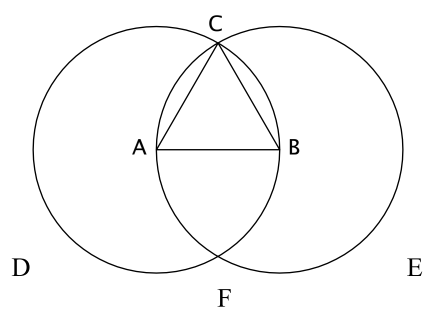
If one likes, this process can now be repeated, with the complex predicates just formed playing the role that simple predicates originally played at the first stage, yielding the new sentences {∃x[Rax→Sxd], ∀x∃y[Sxy&Taya]}.

They can then be conjoined, and the individual constant a substituted for to yield the further one-place complex predicate (in z): ∃x[Rzx→Sxd]&∀x∃y[Sxy&Tzyz].

We can use these procedures to build to the sky, repeating these stages of concept construction by sequential **composition** and **decomposition** as often as we like.
Frege's rules tell us how to compute the inferential roles of the concepts formed at each stage, on the basis of the inferential roles of the raw materials, and the operations applied at that stage.
This is the heaven of concept formation he opened up for us.

Kant's challenge: to understand *multipremise* inferences.
How can a *set* of premises jointly *contain* (in the sense of *imply*) something that is not *contained in* any of the premises individually? Where does the extra content come from?

Diagram of proposition I.1 of Euclid's *Elements*.



Frege: The modes of presentation contained in the sense of one expression both consist of functions-applied-to-arguments into which an expression such as '2^4' can be analyzed and each correspond to a pattern of inferences (implications and incompatibilities) relating the expressions to their substitutional variants.

Bolzano's strategy is to understand the *modes of combination* [*Verbindungsarten*] of expressions—the surplus, above the mereological 'content' that consists in the union of their parts, and which distinguishes 'learned son of an ignorant father' from 'ignorant son of a learned father'—in terms of *relations* of expressions to other expressions. Those relations that functionally define modes of combination are semantic invariances under substitution (his 'variation'), each codifying a pattern of inferences.